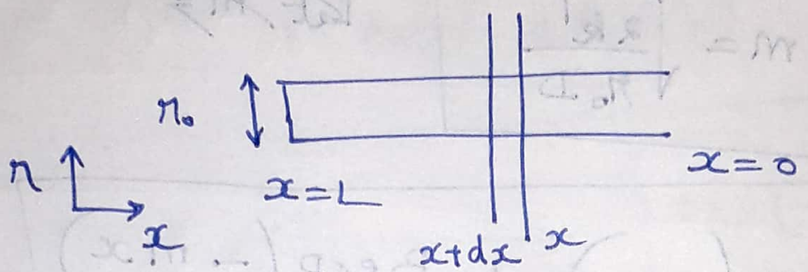
(1) $r' = k' C_s$



Mole Balance

$$\left\{ \text{Rate of Moles in} - \text{Rate Moles out} \pm \text{Rate of generation or consumption} \right\} = 0$$

$$\pi r_0^2 \left(-D \frac{dC_s}{dx}\right)\Big|_x - \pi r_0^2 \left(-D \frac{dC_s}{dx}\right)\Big|_{x+dx} - k' C_s (2\pi r_0)\Delta x = 0$$

÷ by $\pi r_0^2 \Delta x$.

$$\frac{d}{dx}\left(D \frac{dC_s}{dx}\right) = \frac{k' C_s}{D} \frac{2}{r_0}$$

$$\boxed{\frac{d^2 C_s}{dx^2} = \frac{2k' C_s}{D r_0}}$$

$$\left.\begin{aligned} \text{At} \quad & x = 0 ; \ C = C_b \\ & x = L ; \ \frac{dC}{dx}\Big|_{x=L} = 0 \end{aligned}\right\} \text{BCs}$$

$$\boxed{\frac{dC_s}{dx} = A\exp(mx) - B\exp(-mx)}$$

At $x = 0$

$C_s = C_b$

$$\boxed{C_b = A + B}$$

At $x = L$

$$\left.\frac{dC_s}{dx}\right|_{x=L} = 0$$

$$0 = A\exp(mL) - B\exp(-mL)$$

$$\boxed{0 = A - B\frac{\exp(-mL)}{\exp(mL)}}$$

$$\boxed{B = \frac{C_b}{1 + \exp(-2mL)}}$$

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$$\boxed{A = \frac{C_b \exp(-2mL)}{1+\exp(-2mL)}}$$

$$C_s = \frac{C_b \exp(-2mL)\exp(mx)}{1+\exp(-2mL)} + \frac{C_b \exp(-mx)}{1+\exp(-2mL)}$$

$$\boxed{C_s = \frac{C_b}{1+\exp(-2mL)}\left\{\exp(-2mL)\exp(mx) + \exp(-mx)\right\}}$$

Thiele modulus, $\phi = mL$

$$C_s = \frac{C_b}{1+\exp(-2\phi)}\left\{\exp(-2\phi)\exp\left(\frac{\phi x}{L}\right) + \exp\left(-\frac{\phi x}{L}\right)\right\}$$

Multiply & divide by $\exp\phi$

$$= \frac{C_b \exp(\phi)}{\exp(\phi)+\exp(-\phi)}\left\{\exp(-2\phi)\exp\left(\frac{\phi x}{L}\right) + \exp\left(-\frac{\phi x}{L}\right)\right\}$$

$$= \frac{C_b}{\exp(\phi)+\exp(-\phi)}\left\{\exp(-\phi)\exp\left(\frac{\phi x}{L}\right) + \exp(\phi)\exp\left(-\frac{\phi x}{L}\right)\right\}$$

$$= \frac{C_b}{\exp(\phi) + \exp(-\phi)} \left\{ \exp\left(\phi\left(-1 + \frac{x}{L}\right)\right) + \exp\left(\phi\left(1 - \frac{x}{L}\right)\right) \right\}$$

$$C_s = \frac{C_b \left\{ \dfrac{\exp(\phi(1 - x/L)) + \exp(-\phi(1 - x/L))}{2} \right\}}{\left[\dfrac{\exp(\phi) + \exp(-\phi)}{2} \right]}$$

$$\boxed{C_s = \frac{C_b \cosh\left[\phi\left(1 - \frac{x}{L}\right)\right]}{\cosh \phi}}$$

(3)

$$R_{obs} = -D \left.\frac{dc}{dx}\right|_{x=0} (\pi r_0^2)$$

$$= -\frac{D \pi r_0^2 C_b}{\cosh \phi} \left\{ \sinh\left[\phi\left(1 - x/L\right)\right] \right\}_{x=0} \left(-\frac{\phi}{L}\right)$$

$$\boxed{R_{obs} = \frac{D \pi r_0^2 C_b \phi}{L} \tanh \phi}$$

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$$R_{ideal} = k' C_b (2\pi r_o L)$$

~~Effectiveness factor~~

$$\eta = \frac{k' C_b (2\pi r_o L)}{\frac{D \pi r_o^2 C_b \phi \tanh\phi}{L}}$$

$$= \frac{2k' L^2}{D r_o \phi \tanh\phi}$$

$$\neq \phi^2$$

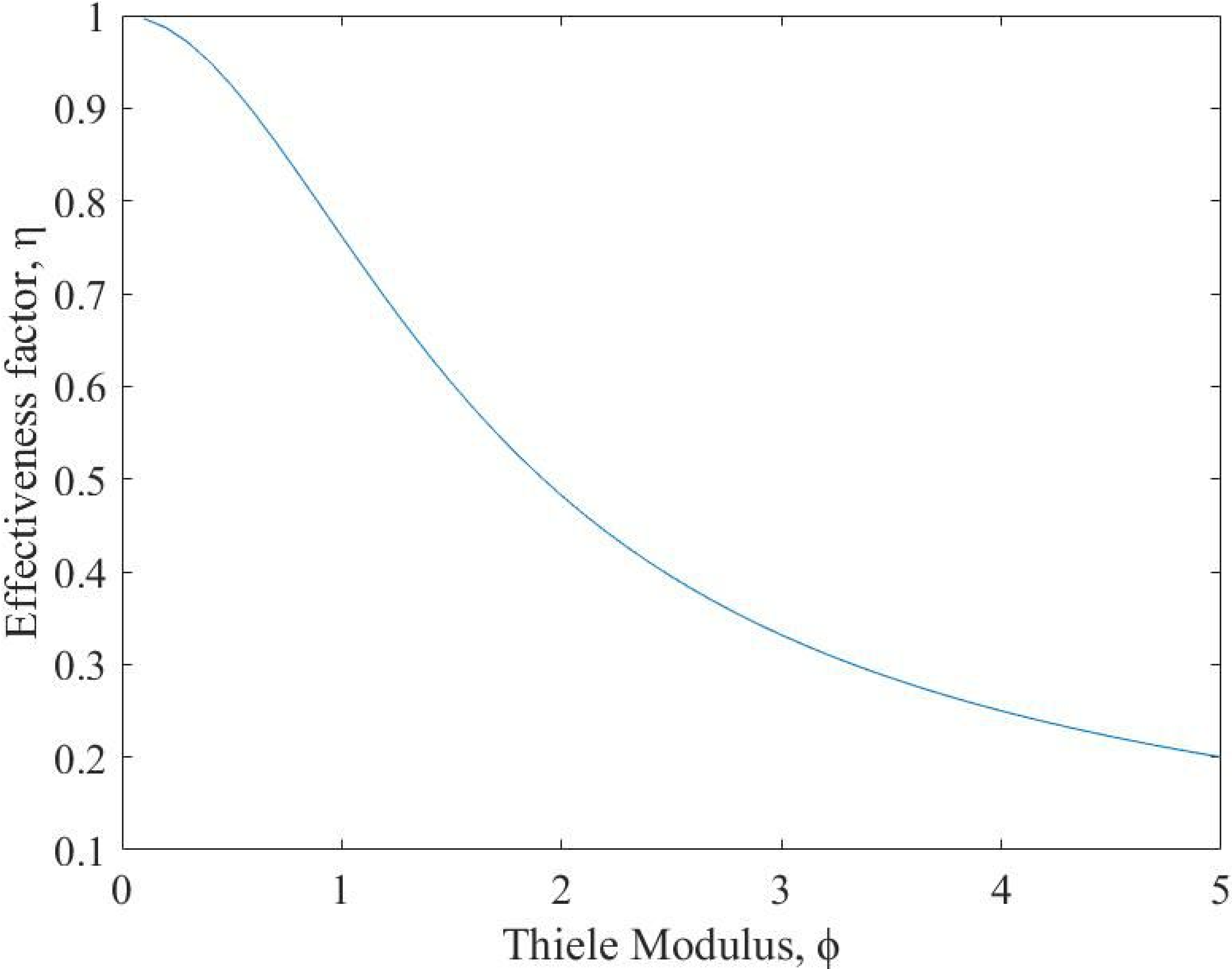
Effectiveness factor

$$\eta = \frac{R_{obs}}{R_{ideal}}$$

$$= \frac{D \pi r_o^2 C_b \phi \tanh\phi}{L k' C_b (2\pi r_o) L}$$

$$= \frac{\phi \tanh\phi}{\frac{2k' L^2}{D r_o}}$$

$$\boxed{\eta = \frac{\tanh\phi}{\phi}}$$



1
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
Effectiveness factor, η
0
1
2
3
4
5
Thiele Modulus, ϕ